Week 100: Distance in nested worlds

Adult guide built from the problem files · Unpiloted; physically unrehearsed.

The answer keys, proofs, preparation and launch are in the print guide, week-100-facilitator.pdf, whose source is guide/facilitator.tex and guide/solutions.tex. This view lists each problem's number, route and idea.

Problems and routes

Problem	3–5	route
Build a world and read its distances	P1	first
Which triples of distances can loops make?	P2	first
Balls around a card	P3	first
Rebuild a world from ten clue cards	P4	first
Repair an impossible set of clues	P5	return
Can exactly one distance be largest?	P6	return
Can balls partly overlap?	P7	return
The eight-number world	P8	return
Why the largest number distance occurs twice	P9	return
Find a secret world with few questions	P10	return

First route. Grades 3–5 P1–P4. **Return visits.** Grades 3–5 P5–P10.

Key, one entry per problem

Build a world and read its distances

Grades 3–5 P1 · first route

Idea. A distance is the label of the smallest common group, and moving cards within their groups keeps every distance.

Which triples of distances can loops make?

Grades 3–5 P2 · first route

Idea. Among any three cards the largest distance occurs at least twice: 2, 2, 4 and 2, 4, 8 cannot be made.

Balls around a card

Grades 3–5 P3 · first route

Idea. Every member of a ball is a center of it, and a radius need not be a label.

Rebuild a world from ten clue cards

Grades 3–5 P4 · first route

Idea. Complete consistent distances determine the groups and their labels.

Repair an impossible set of clues

Grades 3–5 P5 · return route

Idea. One bad triangle proves a table impossible without saying which clue is wrong; exactly three one-clue repairs work.

Can exactly one distance be largest?

Grades 3–5 P6 · return route

Idea. Nested groups prove the largest of three distances occurs at least twice in every world.

Can balls partly overlap?

Grades 3–5 P7 · return route

Idea. Balls that share a card are nested, and any member can be the center.

The eight-number world

Grades 3–5 P8 · return route

Idea. Cards 0-7 with distances 8, 4, 2 by parity and remainders mod 4 form a nested world (a scaled 2-adic distance).

Why the largest number distance occurs twice

Grades 3–5 P9 · return route

Idea. If 2^k divides $a - b$ and $b - c$, it divides $a - c$.

Find a secret world with few questions

Grades 3–5 P10 · return route

Idea. Asking all 15 pairs always settles a six-card world, and when every answer is 8 nothing less can.